18 13 1

→ Queue → D.S FCFS
First Come First Serve

Push
Capacity = 5

0	1	2	3	4
18	13	8	20	

front, rear

```
front = -1
rear = -1
capacity = 5;
```

```
Class Queue
{
    public
    int front, rear, capacity;
    int * arr;

    Queue (int size)
    {
        front = rear = -1
        arr = new int [size]
        capacity = size;
```

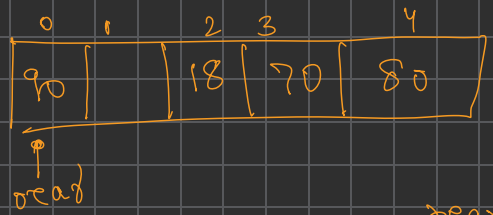
```
bool isEmpty()
{
    return rear == -1;
}

bool isFull()
{
    return rear == capacity-1;
}

void push(int x)
{
    if (isEmpty())
    {
        front = rear = 0;
        arr[rear] = x;
        return;
    }
    elseif (isFull())
    {
        cout << "Queue is full";
        return;
```

```
}
else
{
    rear = rear+1;
    arr[rear] = x;
}


void pop()
{
    if (isEmpty())
    {
        cout<<"Queue is Empty";
        return;
    }
    else
    {
        front = front+1;
        if (front>rear)
            front = rear=-1;
    }
}
```



front = -1

rear = -1

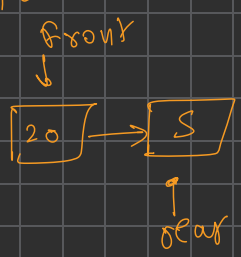
capacity = 5

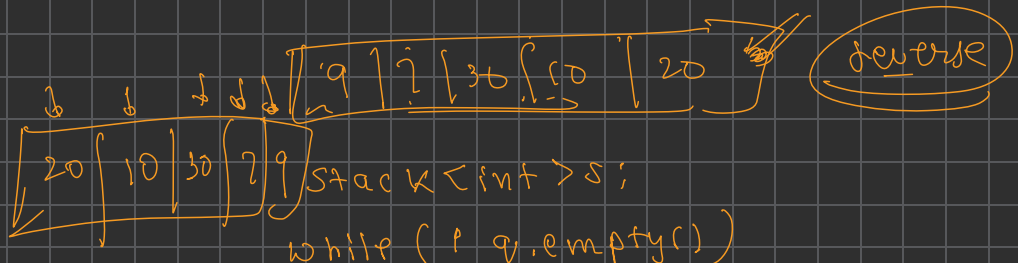
totalElement = 4

rear = (rear + 1) % capacity

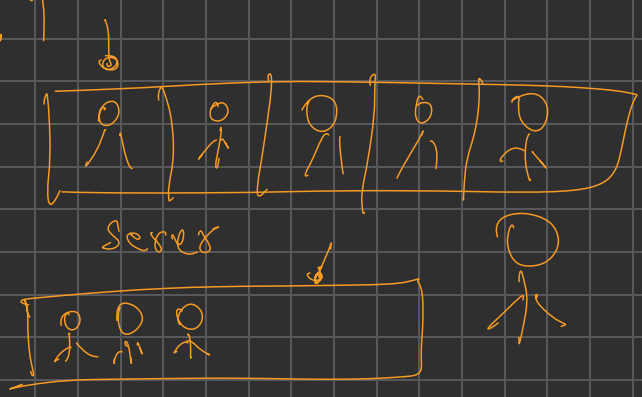
front = (front + 1) % capacity

Linked List :






```
stack<int> s;
while (!q.empty())
{
    s.push(q.front());
    q.pop();
}
while(!s.empty())
{
    q.push(s.top());
    s.pop();
}
```



5

$k = 4$

$n = 9$

5 6 4 8 13 17 18 11 19

5 6 4 8 13 17 18 11 19

$2 - 1 = 1$

Answer = 1 sec

0	1	2	3
2	3	2	4

$k = 2$

$2 + 2 + 2 + 1 = 7$ ✓ 31

1 sec

1 sec

1 sec

1 sec

1 sec

1 sec

1 sec

1 sec

[100 500 300 200 70

910 240]

199 Airlief

100 + 200 + 200 + 200 + 70 + 199 + 199

Total tiy

Person

8 items 8 - 1 = 7

10	6	8	9	5	12

= 41 sec cov

Total time: 8 + 6 + 8 + 7 + 5 + 7

1 4 3

4 3 1

3 1 3

X 3 2